

Angqi Li

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Education

Michigan State University , Ph.D. in Computational Mathematics, and Science Engineering	Jan 2023 – Present
Michigan State University , MS in Computational Mathematics, and Science Engineering	Jan 2023 – Dec 2025
Michigan State University , B.S. in Chemical Engineering	Sept 2018 – May 2022
• Minor: Computer Science & Mathematics	

Experiences

Research Assistant , Michigan State University	Jan 2023 – Present
• Architected and deployed a suite of advanced deep learning models, including Transformers, Diffusion Models, and Deep Image Priors, to analyze natural and medical images. • Spearheaded a key project component focused on model development, leading to more robust data-driven insights for brain signal decoding. • Played an instrumental role in writing and securing a research grant, ensuring the continuation and expansion of our work. • Contributed to a cross-functional team of scientists and engineers to advance brain signal decoding research.	
Laboratory Assistant , Shanghai Jiaotong University – Shanghai, China	Apr 2021 – Aug 2021
• Assisted 3 researchers in analyzing data for 4 projects, debugging prototypes. • Conducted 2 independent experiments, improving tissue clearing (CUBIC) efficiency by 10%. • Played a key role in developing new non-toxic, organic transparency methods as part of the core research team.	

Projects

DiffuseClean - A Diffusion Model for EEG Artifact Removal
• Built a diffusion model to remove ocular artifacts, improving EEG signal quality. • Achieved 10% lower RRMSE and 5% higher Cosine Similarity, outperforming traditional methods.
Fine-Tuning a Large Language Model
• Trained and fine-tuned a Large Language Model (LLM) for domain-specific applications using PyTorch and Hugging Face Transformers. • Deploying the model for chatbots and automated summarization.

Activities

Organizer and Co-host , Data-driven Understanding of Meditation and Consciousness (DUNES) Webinar Series	May 2024 - Present
• Expanded global collaboration by organizing and co-hosting a webinar series on Zoom. • Engaged 300 researchers and students from 5 countries, fostering interdisciplinary dialogue.	

Skills

• Programming Languages: Python, C++ , C, R, MATLAB, SQL, JavaScript
• Machine Learning: Data Preparation, Feature Engineering, Model Optimization, ML Framework
• Data Science Tools: Jupyter Notebooks, TensorFlow, PyTorch, Scikit-learn, Docker, AWS

Publications and Conferences

- **Scan-Adaptive MRI Undersampling Using Neighbor-Based Optimization (SUNO)** IEEE Transactions on Computational Imaging 2026
Siddhant Gautam, *Angqi Li*, Nicole Seiberlich, Jeffrey A. Fessler, Saiprasad Ravishankar
10.1109/TCI.2026.3653330
- **Not all mantra meditations are equal: Emergence of divergent alpha oscillatory dynamics across mantras** bioRxiv 2026
Angqi Li, Julio Rodriguez-Larios, Mengsen Zhang, Taosheng Liu, Barry H Cohen, Saiprasad Ravishankar
<https://doi.org/10.64898/2026.02.26.707862>
- **L-FAME: Longitudinal Focused Attention Meditation EEG Dataset and Benchmark** arXiv 2026
Angqi Li, Ab Basit Rafi Syed, Hamzeh Alzweri, Taosheng Liu, Barry H Cohen, Saiprasad Ravishankar
<https://doi.org/10.48550/arXiv.2605.22893>
- **Quantifying Improvements in Cognitive Skills, Stress, and Mindfulness from Mantra-based and Breath-focus Meditation techniques** Neuroscience 2025 San Diego
Angqi Li, Ab B. Syed, Krishna Ika, Mengsen Zhang, Barry H. Cohen, Saiprasad Ravishankar
- **Meditation-journey: Dataset and Benchmarks for Longitudinal Study of Different Meditation Techniques** Neuroscience 2025 San Diego
Angqi Li, Ab B. Syed, Mengsen Zhang, Barry H. Cohen, Saiprasad Ravishankar
- **Analysis of Longitudinal Variations in Brain Oscillations for Breath-Focus and Mantra-Based Meditation** Neuroscience 2025 San Diego
Angqi Li, Mengsen Zhang, Barry H. Cohen, Saiprasad Ravishankar
- **Patient-Adaptive and Learned MRI Data Undersampling Using Neighborhood Clustering** ICASSP 2024
Siddhant Gautam, *Angqi Li*, Saiprasad Ravishankar
ICASSP48485.2024.10446528
- **Understanding Longitudinal Effects of Mantra Meditation and Breath-focused Meditation using EEG** Neuroscience 2024 Chicago
Angqi Li, Pratham Pradhan, Krishna Ika, Mengsen Zhang, Barry H. Cohen, Saiprasad Ravishankar
PSTR421.18/P3
- **Patient-adaptive and Learned MRI Data Undersampling Using Neighborhood Clustering** NeurIPS 2023
Siddhant Gautam, *Angqi Li*, Evan Bell, Saiprasad Ravishankar
Poster link
- **Measuring the Effectiveness of Mantra-based Meditation using EEG Data Analysis** Neuroscience 2023 Washington, D.C.
Angqi Li, Pratham Pradhan, Annie Wozniak, Krishna Ika, Barry H. Cohen, Saiprasad Ravishankar
PSTR564.01/SS3